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Forecast revisions of Mexican inflation and GDP growth

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ABSTRACT

We analyze the forecasts of inflation and GDP growth contained in the Banco de México's Survey of Professional Forecasters for the period 1995–2009. The forecasts are for the current and the following year, and comprise an unbalanced three-dimensional panel with multiple individual forecasters, target years, and forecast horizons. The fixed-event nature of the forecasts enables us to examine their efficiency by looking at the revision process. The panel structure allows us to control for aggregate shocks and to construct a measure of the news that impacted expectations in the period under study. We find that respondents anchor to their initial forecasts, updating their revisions smoothly as they receive more information. In addition, they do not seem to use publicly-known information in an efficient manner. These inefficiencies suggest clear areas of opportunity for improving the accuracy of the forecasts, for instance by taking into account the positive autocorrelation found in forecast revisions.
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1. Introduction

It has become common practice in several central banks and other institutions to design and collect surveys containing professionals' forecasts of a range of macroeconomic and financial variables. These surveys can be particularly helpful for understanding the expectation formation mechanism of private agents. Equally importantly, they can also serve as an input in a number of decision-making processes, for instance monetary policy. Given their importance, it is natural to ask about their accuracy and efficacy.

In this paper, we analyze the forecasts of inflation and GDP growth supplied by the individual respondents to the Survey of Professional Forecasters (SPF) which has been conducted monthly by the Banco de México since 1995.[1] These forecasts comprise a three-dimensional panel dataset, with the additional dimension arising from the collection of forecasts at several horizons (Davies & Lahiri, 1995). The forecasts that we use are focused on the year-end outcome in the year in which the survey is collected, and in the following year. The forecasts are revised in response to new information which arises between surveys, and eventually form a sequence of 24 forecasts before the respective outcome is known. This type of expectation is commonly referred to as a fixed-event forecast. Following Isiklar and Lahiri (2007), we use this real-time survey data to analyze the evolution of these fixed-event forecasts over various horizons.

We study the way in which information is incorporated into these forecasts. If it is incorporated efficiently, forecast errors and revisions should not be predictable (Nordhaus, 1987). When the information incorporated is

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[1] The survey is called Encuesta sobre Expectativas de los Especialistas en Economía del Sector Privado. Banco de México is México's central bank, with webpage http://www.banxico.org.mx.

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publicly available data that might have been employed in the construction of previous forecasts, the forecasts are said to satisfy the property of strong efficiency. If the information used is past forecast errors or revisions, then the forecasts are said to satisfy the property of weak efficiency. Note that the latter is a necessary condition for the former, but it is not sufficient.

The fixed-event nature of the forecasts enables us to examine the forecast efficiency by looking at forecast revisions. Rationality tests based on forecast revisions are attractive because they are not sensitive to the data generating process or to data revisions. This is important for Mexican data, given that Mexico adopted an inflation targeting regime in 2001, with inflation apparently changing from a very persistent process to a stationary one around that year (Chiquiar, Noriega, & Ramos-Francia, 2010).[2] In the case of GDP, it had a major revision in 2008, in which the base year was changed from 1993 to 2003.[3] Two other advantages of using the fixed-event forecast in the SPF are that we have forecasts for reasonably long horizons (up to twenty-four months), and that these are revised on a monthly basis.[4]

The panel structure of the data makes it possible to separate forecast errors into macroeconomic aggregate shocks and forecaster-specific idiosyncratic errors (Davies & Lahiri, 1995). The aggregate shocks constitute a measure of the news that impacted GDP growth and inflation expectations in the period under study. Following Davies and Lahiri (1995) and Boero, Smith, and Wallis (2008b), we calculate a measure of the volatility of the aggregate shocks for the Mexican case. In addition, in order to take full advantage of the panel structure of the data, we consider pooling fixed-event forecasts across events and over individual respondents in order to deliver more powerful tests of forecast efficiency (Clements, 1997; Keane & Runkle, 1990).

We find that SPF respondents seem to start with a fixed value for the initial forecast, around 4.1% for inflation (during the inflation targeting period) and around 3.8% for GDP growth. Compared to the event's realization, these values appear to be too rosy, and hence these longer-run forecasts tend to under-predict inflation and over-predict GDP. The forecasters appear to start incorporating news into the forecasts around 12 and 16 months before the actual realizations of the end-of-the-year inflation and real GDP growth, respectively. We also find that forecasters anchor to their initial forecasts, updating their revisions smoothly as they receive more information. Further results indicate the presence of inefficiencies in forecasters' use of information about the past evolution of monthly inflation and monthly measures of economic activity. All of these inefficiencies suggest that there are clear areas of opportunity for significant improvements in the accuracy of the forecasts. For instance, the positive autocorrelations found in forecast revisions could be used to predict subsequent revisions.

With respect to the aggregate shocks, we find high levels of volatility for both inflation and GDP growth around the crisis of 1994–1995. For the latter, we also find high levels of volatility around the recent global financial crisis. An important reduction in inflation uncertainty is evident following Banco de México's implementation of an inflation targeting framework. There is also a cluster of positive aggregate news to inflation around that time. Finally, GDP growth uncertainty appears to have a cyclical component.

The paper proceeds as follows. Section 2 provides information about the SPF and the data employed, and presents an analysis of the information content of the forecasts. Section 3 derives the efficiency tests used in this paper. Section 4 presents the results of the forecast efficiency tests. The estimates of the aggregate shocks and their volatility appear in Section 5. Finally, Section 6 discusses the results and concludes.

2. Data

2.1. Banco de México's survey of professional forecasters

Banco de México has been conducting the SPF on a monthly basis since September 1994. Nowadays, the SPF covers around 20 macroeconomic variables related to investment, production, labor markets, public finance and international trade. In addition, the survey also asks the professional forecasters for their views on some qualitative aspects of the Mexican economic environment.

The numbers of forecasters in the survey have varied over the years, with fewer than 15 respondents in the earlier months, and approximately 30 regular respondents in recent surveys. The specialists who participate in the survey come mainly from commercial banks and other financial institutions (57%), followed by consulting firms (29%) and industrial and academic institutions (14%) in smaller proportions. Their forecasts are gathered by mail in the second half of each month, and the un-weighted mean (consensus forecast) is published monthly by the Banco de México in a detailed report that contains the evolution of these expectations.

In this paper, we analyze the forecasts collected in the SPF that focus on predicting the current and following year of annual CPI inflation and the average GDP growth over the year, for a sample that runs from as early as January (for inflation) and March (for GDP growth) of 1995 to December 2009. Ideally, given the frequency of the survey and the structure of the data, we should be able to extract a sequence of 12 fixed-event forecasts for each forecaster for the target year 1995 (10 forecasts for GDP growth), 24 fixed-event forecasts for the target years 1996 to 2009 (22 for GDP growth in 1996), and 12 forecasts for the year

[2] Mexico took the first step towards inflation targeting in 1999 and consolidated it in 2001. Starting in 2003, the long-term inflation objective for the CPI was set at 3%, with a variability interval of plus/minus 1%. See Banco de México's Monetary Program for 2003.

[3] The revision was announced in April of 2008. It included, among other things, increasing the number of activities considered from 362 to 750. See http://www.inegi.org.mx.

[4] Despite these advantages, the bulk of the forecast evaluation literature has focused on fixed-horizon forecasts, which are a series of predictions for different events at a fixed horizon. See for example Mincer and Zarnowitz (1969) for early work on fixed-horizon forecasts, and Capistrán and López-Moctezuma (2010) for an evaluation of fixed-horizon predictions from Banco de México's SPF.

2010, with $h = 23, 22, \ldots, 1, 0$; where horizon 23 corresponds to the forecast made in January of a given year for the outcome of the following year, and horizon 0 corresponds to the forecast released in December of a given year for the outcome of the current year. However, in practice, the sequences of individual fixed event forecasts take the form of an unbalanced panel for two reasons. First, individual forecasters enter and exit the survey frequently, generating a source of missing data (see Capistrán & Timmermann, 2009b). Second, in its first three years, the SPF did not ask systematically for next-year inflation and GDP growth forecasts, causing additional gaps in the dataset.

To avoid the complications caused by long gaps in the data, our main analysis refers to those respondents who reported longer spans of consecutive forecasts using the following metric. For each forecaster, we calculated the numbers of observations with four or more consecutive forecasts, and selected those forecasters with at least the median number of observations.[5] Based on this metric, our empirical analyses focus on a sub-sample of 41 forecasters.

2.2. Descriptive statistics

As a first step in the exploratory data analysis of the forecasting process in Mexican inflation and GDP growth, Figs. 1 and 2 present a graphical summary of the cross-section mean and standard deviation of the panel forecasts for each year under study, together with the available actual outcomes. In these figures, the sample mean is the common choice of a "consensus" forecast, and the standard deviation across forecasters is a measure of the disagreement among respondents and could be viewed as a proxy for the uncertainty surrounding panel members' responses about future outcomes (Harvey, Leybourne, & Newbold, 2001). It might be expected that the variability in panel members' forecasts would decrease as they approach the outcome of interest. This might happen because the forecasters should provide more accurate estimates as new information becomes available (Isiklar & Lahiri, 2007).

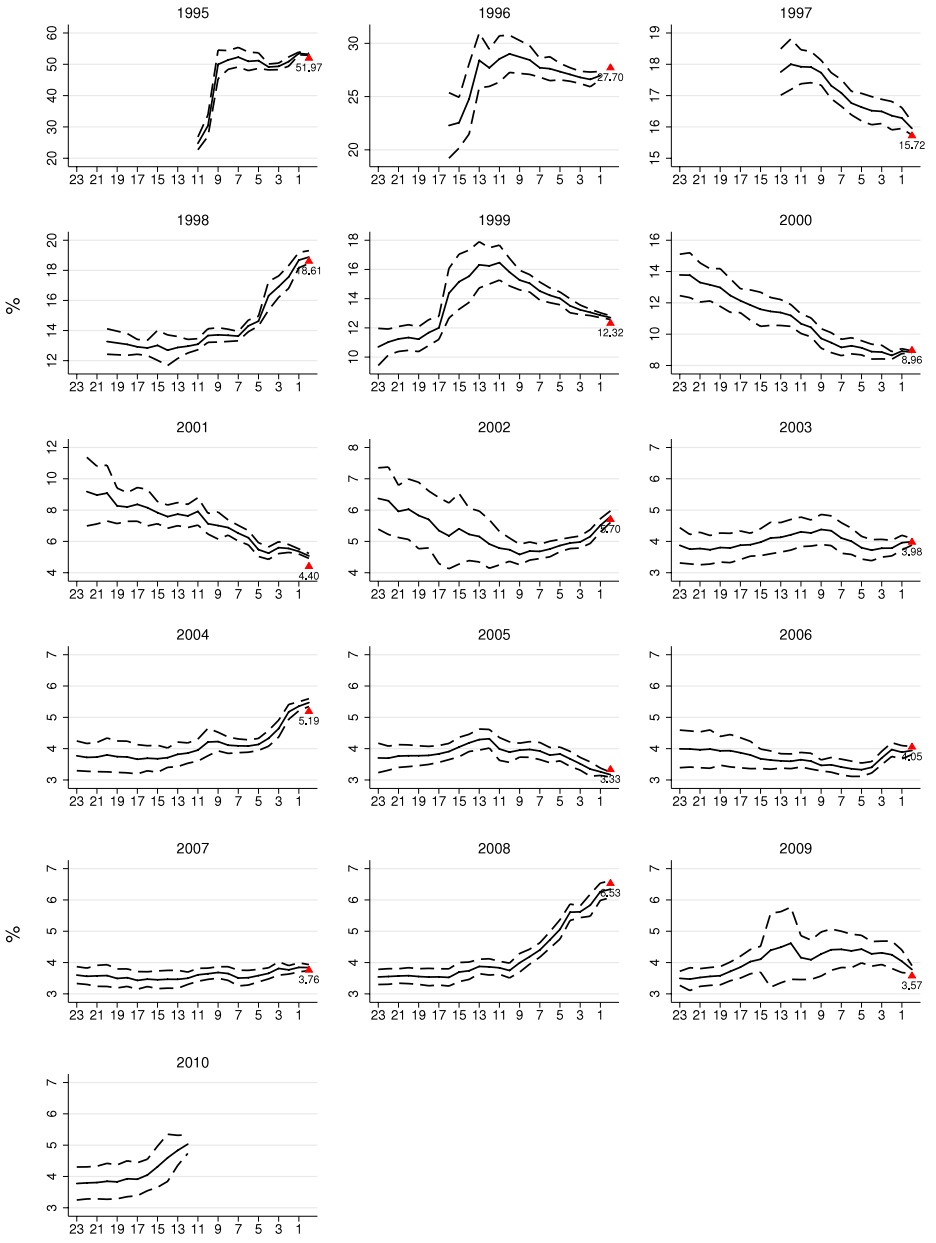
The general picture is that the evolution of the mean forecast seems to converge towards the actual outcome, which is not surprising, since, for shorter horizons, much of the actual outcome has already been published, although the speed of convergence seems to vary across years and across the variables being forecasted.

Similarly to the majority of the developed countries examined by Isiklar and Lahiri (2007), the initial average forecasts produced by the respondents of the SPF, namely the 24-month-ahead forecasts, seem to start from very similar initial points. This happens regardless of their final expectations, and thus, of the actual value of the variables of interest, as can be seen in Table 1. Isiklar and Lahiri (2007) conjectured that the long term forecasts may be regarded as the unconditional mean of the time series of interest. Lahiri and Sheng (2008, 2010) find that almost all of the variance in the initial forecasts can be explained by the prior beliefs of the forecasters, which in turn generates the anchoring behavior we also observe in the Mexican data. In the case of inflation forecasts, the initial expectations seem to change considerably between 1999 and 2001, which coincides with a strong disinflation effort and the transition towards an inflation targeting regime. From 2002 onwards, the longer-run expectations are located at around 4%. This number is the upper limit of the variability band around the inflation target announced by Banco de México of 3% plus/minus 1%. Comparing this initial forecast with the average inflation at the end of the year for the period 2003–2009, 4.5%, there appears to be a systematic bias of the initial forecasts towards under-predicting inflation (the mean error for this period is 0.5%). In the case of the GDP growth forecasts, the third column of Table 1 shows that the initial forecasts start from between 3.3% and 4.6% for the whole period 2000–2009, with a mean value of 3.8%. Given that the average GDP growth over this period is 1.9%, the initial predictions of the forecasters seem to over-predict GDP growth, on average, by 1.9%. This systematic bias towards under-predicting inflation and over-predicting GDP growth in the last part of the sample is evident in Fig. 3, which shows the evolution of the mean forecast error, defined as the actual value of the variable of interest minus the forecast, against the forecast horizon. As can be seen from these charts, there is a systematic bias across horizons for both inflation and GDP growth forecasts that reduces as the forecast horizon increases.

Another relevant feature of the evidence presented above is that the sequences of consensus forecasts seem to move steadily up or down towards the actual outcomes. At first glance, this feature seems inconsistent with the behavior of an optimal forecast, which should look more like a white-noise process, since optimal forecast revisions should be unpredictable in every period (as is discussed further in the next section). Instead, this pattern appears to follow the logic of respondents sticking to their past forecasts as new information arises. In the next section, we develop the tools necessary to test whether this pattern is compatible with the efficient use of information.

Finally, it is also possible to see from Figs. 1 and 2 that, for the majority of the events analyzed in the period 1999–2009 (for which the SPF includes 24-month-ahead forecasts), the cross-section average forecast does not seem to change considerably for approximately the first 6 to 8 months, which seems to indicate that the initial months do not convey sufficient relevant news to persuade the forecasters to revise their predictions systematically. This evidence is also consistent with that reported by Isiklar and Lahiri (2007) for developed countries. Furthermore, Fig. 4 shows the behavior of the Diebold and Kilian (2001) statistic for measuring the predictability of a variable. This measure is $pred_{h,23} = 100\left(1 - \frac{MSE_h}{MSE_{23}}\right)$, where MSE_h is the mean squared error of the forecasts at horizon h (the MSE is taken across events). This statistic shows the evolution of the information content of the forecasts, as measured by the decrease in MSE over that of the 24-month-ahead forecast (the forecast at horizon 23, in our notation). For inflation, we see that the predictability

[5] The results are robust to changes in the threshold of consecutive forecasts and to alternative metrics based on the total number of observations for each forecaster (results are available upon request).



* The horizontal axis plots the forecast horizon when the prediction was produced.

Horizon 23 corresponds to the forecast made in January of a given year for the outcome of the following year.

Horizon 0 corresponds to the forecast released in December of a given year for the outcome of the current year.

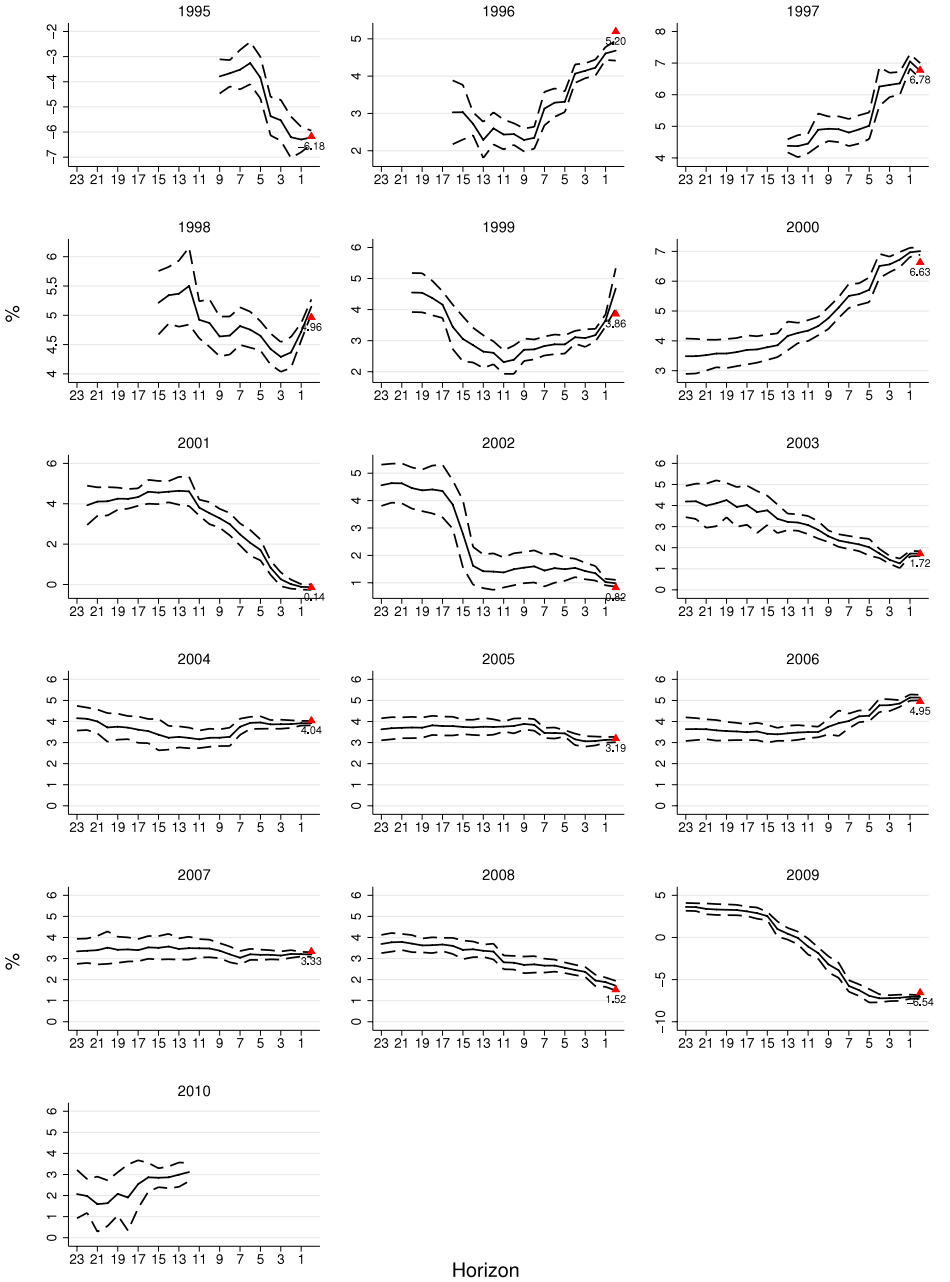
** The red △ denotes the actual outcome.

*** Dashed lines correspond to the mean forecast +/- the sample standard deviation.

Fig. 1. Annual inflation forecasts and cross-section dispersion, 1995–2010.

first drops, then increases for about four months, remains about the same for another six, and finally increases steadily for the next eleven months. The deterioration in forecast performance up to horizon 12 can be explained by the large initial bias of some forecasters, particularly for the year 2001. At horizons 22 and 21, the consensus forecast was located at around 9%, with some forecasters predicting an annual level of inflation as high as 18%, when in fact the actual level of inflation that year ended up around 4.40%.[6]

[6] See Capistrán (2007) for more on the multi-horizon properties of rational forecasts and a test of rationality based on the non-increasing property of MSEs as the forecast horizon decreases.



* The horizontal axis plots the forecast horizon when the prediction was produced.
Horizon 23 corresponds to the forecast made in January of a given year for the outcome of the following year.
Horizon 0 corresponds to the forecast released in December of a given year for the outcome of the current year.
** The red △ denotes the actual outcome.
*** Dashed lines correspond to the mean forecast +/- the sample standard deviation.

Fig. 2. Annual GDP growth forecasts and cross-section dispersion, 1995–2010.

The behavior of the shorter-horizon forecasts is consistent with rapid gains in predictability, as the slope of the $pred_{h,23}$ measure is very steep. For GDP growth, we see that each additional month increases the information content of the forecasts, as the MSEs improve steadily over those of previous months. There is a change in the slope starting at horizon 16, indicating large gains in predictability starting about four quarters ahead of the realization of the event.

3. Efficiency tests

Efficiency tests for fixed-event forecasts using forecast revisions were first introduced by Nordhaus (1987), who

Table 1
Initial forecasts vs. actual values, inflation and GDP growth.

Event	Inflation		GDP growth	
	Initial forecast*	Actual value	Initial forecast*	Actual value
1999	10.70	12.32	–	–
2000	13.78	8.96	3.49	6.63
2001	9.18	4.40	3.92	−0.14
2002	6.37	5.70	4.56	0.82
2003	3.88	3.98	4.19	1.72
2004	3.77	5.19	4.16	4.04
2005	3.70	3.33	3.63	3.19
2006	3.99	4.05	3.64	4.95
2007	3.60	3.76	3.34	3.33
2008	3.54	6.53	3.69	1.52
2009	3.50	3.57	3.63	−6.54
Mean**	4.04	4.51	3.83	1.95
Standard deviation**	0.95	1.15	0.38	3.60

* The initial forecast is the cross-section mean forecast made at horizon 23 (i.e., a 24-month-ahead forecast). ** The mean and standard deviations include the periods 2002–2009 for inflation forecasts and 2000–2009 for GDP growth forecasts.

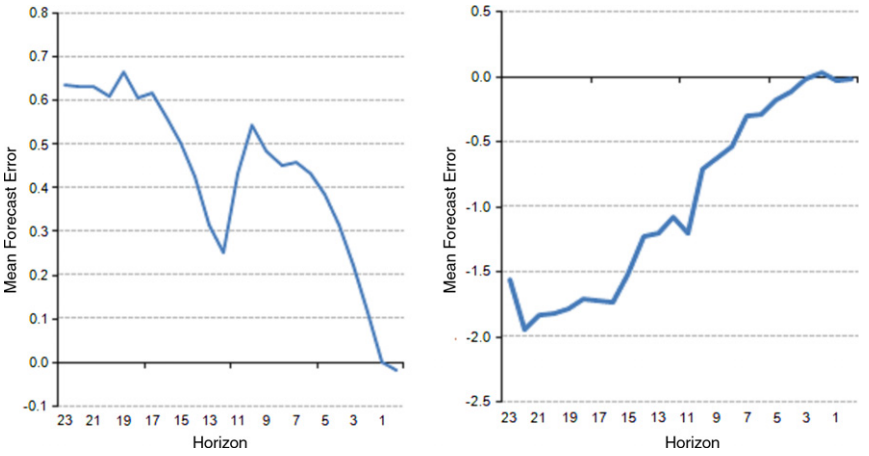


Fig. 3. Average forecast error by horizon.

discussed the efficiency of forecast revisions for a single terminal event. Later on, Davies and Lahiri (1995) considered exploiting the panel data structure of survey forecasts under an econometric framework that renders it possible to decompose forecasts errors into macroeconomic aggregate shocks and forecaster-specific idiosyncratic errors.[7] Forecast revisions of professional forecasters in developed economies have been analyzed extensively under this framework.[8] However, to the best of our knowledge, no previous study has analyzed the efficiency of fixed-event expectations in survey data in the case of developing economies.

First, we derive the properties of optimal fixed event forecasts under a general loss function. Then, we use the quadratic loss function to derive explicit tests that involve the forecast revisions. To facilitate the exposition, the optimal properties are derived for a representative forecaster. Later, we show the advantages of having a panel of forecasters.

3.1. Optimal forecasts

We are interested in forecasting the outcome of an event that is going to be realized at period τ, y_τ, with information up to h periods before. The information set is denoted $\boldsymbol{I}_{\tau-h}$, and contains at least the realizations of the variable to be forecasted, and possibly also lagged values of other variables, up to period $\tau - h$.

The optimal forecast computed at period $\tau - h$ conditional on $\boldsymbol{I}_{\tau-h}$ is defined as

$$f^*_{\tau,h} \equiv \arg\min_{f_{\tau,h}} E\left[L(y_\tau - f_{\tau,h}) \mid \boldsymbol{I}_{\tau-h}\right],$$

[7] See Davies, Lahiri, and Sheng (2011) for a recent summary of the literature analyzing three-dimensional panels of forecasts.

[8] For US data, see Nordhaus (1987) and Swidler and Ketcher (1990). Bakhshi, Kapetanios, and Yates (2005), Clements (1997), Harvey et al. (2001), and Boero, Smith, and Wallis (2008a); Boero et al. (2008b) analyze UK data. Dovern and Weisser (2008) use data for the G7 countries.

where $L(\cdot)$ is the loss function, and $f_{\tau,h}$ is the forecast for τ made at $\tau - h$. The first-order condition is

$$E\left[L'(y_\tau - f^*_{\tau,h}) \mid \boldsymbol{I}_{\tau-h}\right] = 0, \tag{1}$$

where $L'(\cdot)$ denotes the derivative of the loss function with respect to the predictor, $f_{\tau,h}$.[9] Following Granger (1999) and Patton and Timmermann (2007), this derivative is called the generalized error. It gives the change in total loss which results from a one-unit change in the forecast. Under certain regularity conditions, the first-order condition will be sufficient to give the optimal forecast.

The optimal forecast made at $\tau - (h+1)$, that is, one period before h, has to satisfy

$$E\left[L'(y_\tau - f^*_{\tau,h+1}) \mid \boldsymbol{I}_{\tau-(h+1)}\right] = 0. \tag{2}$$

Assuming that the information set is a filtration, the first difference of the generalized error (times -1, a simple normalization) satisfies

$$E\left[L'(y_\tau - f^*_{\tau,h+1}) - L'(y_\tau - f^*_{\tau,h}) \mid \boldsymbol{I}_{\tau-(h+1)}\right] = 0. \tag{3}$$

By the law of iterated expectations, for any finite function of a random variable belonging to the information set, $\boldsymbol{\zeta}_{\tau-(h+1)} \subset \boldsymbol{I}_{\tau-(h+1)}$, the first difference of the generalized error satisfies the orthogonality condition

$$E\left[(L'(y_\tau - f^*_{\tau,h+1}) - L'(y_\tau - f^*_{\tau,h}))\boldsymbol{\zeta}_{\tau-(h+1)}\right] = \mathbf{0}. \tag{4}$$

Stinchcombe and White (1998) refer to $\boldsymbol{\zeta}$ as the test function. For a given test function, we will construct a test that exploits the orthogonality condition.

3.2. Quadratic loss

3.2.1. A representative forecaster

Under quadratic loss, $L(y_\tau - f_{\tau,h}) = \frac{1}{2}(y_\tau - f_{\tau,h})^2$, the generalized error $L'(y_\tau - f_{\tau,h})$ is identical to the forecasting error $y_\tau - f_{\tau,h}$, and hence, Eq. (1) becomes

$$E\left[(y_\tau - f^*_{\tau,h}) \mid \boldsymbol{I}_{\tau-h}\right] = 0, \tag{5}$$

and we obtain the traditional result that, under quadratic loss, the optimal forecast is the expected value of the variable of interest conditional on the forecaster's information set (Granger & Newbold, 1986). Under this loss function, Eq. (4) becomes

$$E\left[\left(f^*_{\tau,h} - f^*_{\tau,h+1}\right)\boldsymbol{\zeta}_{\tau-(h+1)}\right] = \mathbf{0}, \tag{6}$$

therefore, under optimality, the forecast revision, $\upsilon_{\tau,h} \equiv f_{\tau,h} - f_{\tau,h+1}$, satisfies

$$E\left[(\upsilon^*_{\tau,h})\boldsymbol{\zeta}_{\tau-(h+1)}\right] = \mathbf{0}. \tag{7}$$

From Eq. (7) we can obtain the usual properties of optimal revisions (as per Nordhaus, 1987):

1. $E\left[\upsilon^*_{\tau,h}\right] = 0$, which is obtained with $\boldsymbol{\zeta}_{\tau-(h+1)} = 1$.
2. $E\left[(\upsilon^*_{\tau,h})(\upsilon^*_{\tau,h+j})\right] = 0$, for all $j \geq 1$, which is obtained when $\boldsymbol{\zeta}_{\tau-(h+1)} = \upsilon^*_{\tau,h+j}$.
3. $E\left[(\upsilon^*_{\tau,h})(x_{\tau-(h+1)} - x_{\tau-(h+2)})\right] = 0$, which is obtained when $\boldsymbol{\zeta}_{\tau-(h+1)} = x_{\tau-(h+1)} - x_{\tau-(h+2)}$. In this case, $x_{\tau-(h+1)}$ denotes the most recently observed value of a variable included in the forecaster's information set, and thus, $x_{\tau-(h+1)} - x_{\tau-(h+2)}$ approximates the period-by-period news in this variable.

A series of revisions that satisfy the first two properties are said to be weakly efficient. Property 1 is known as the unbiasedness condition, and implies that there should not be a systematic bias in the revisions. Property 2 implies that the revisions should not be serially correlated. Hence, if the revisions are correlated, it would be evidence of an inefficient use of information under the assumption of quadratic loss. Property 3 is associated with the concept of strong efficiency, as the forecaster's information set includes additional variables other than the past forecast revisions. From the point of view of the researcher, testing this property has the disadvantage, with respect to the first two, that it requires the extra assumption that the variable $x_{\tau-(h+1)}$ actually belonged to the forecaster's information set at the time when the forecast was computed.

Property 2 has been disputed by Davies and Lahiri (1995), who postulate that efficient forecast revision sequences should behave as a first-order moving average, with a negative first-order autocorrelation coefficient, but with all the other autocorrelations equal to zero. However, under our derivation we obtain Nordhaus' (1987) result that optimal forecast revisions (under quadratic loss) should behave as a zero-order moving average.

3.2.2. A panel of N forecasters

In a panel data setting with N forecasters, we can think of the optimal revision for each forecaster as a component term that can be divided into an aggregate shock, $u_{\tau,h}$ (common across forecasters), and an idiosyncratic (white-noise) error, $\varepsilon_{i,\tau,h}$ (specific to each forecaster). As long as $E\left[u_{\tau,h} + \varepsilon_{i,\tau,h}\right] = 0$, the optimal forecast revision can be written as

$$\upsilon^*_{i,\tau,h} = u_{\tau,h} + \varepsilon_{i,\tau,h}. \tag{8}$$

In this context, an estimate of the aggregate shock at each time period is readily available as

$$\widehat{u}_{\tau,h} = N^{-1}\sum_{i=1}^{N} \upsilon^*_{i,\tau,h}. \tag{9}$$

Hence, an estimate of the systematic portion of the forecast errors can be extracted from Eq. (9) assuming: (i) a homogenous quadratic loss across forecasters; and (ii) that each forecaster included to extract the aggregate shock ought to be rational.

The error structure of Eq. (8) implies a three-way panel with multiple individuals, forecast horizons, and events of interest. Davies and Lahiri (1995) have an expression similar to Eq. (8), but they add a bias term and a $MA(1)$ structure in the revision process. These additional terms do not appear in our expression because we have derived it under optimality and using a quadratic loss.

3.3. Tests based on forecast revisions

Following the three-dimensional panel structure of our data, we base our analysis of the properties of efficient forecast revisions, under quadratic loss, on a regression framework that facilitates the pooling of the forecast

[9] We have assumed that integration and differentiation can be interchanged.

revision sequences across different target years and over individual respondents. The general regression for testing efficiency with the forecast revision as the dependent variable can be written as

$$\upsilon_{i,\tau,h} = \boldsymbol{\zeta}'_{\tau-(h+1)}\boldsymbol{\alpha} + \epsilon_{i,\tau,h}, \tag{10}$$
$$H_0 : \boldsymbol{\alpha} = \mathbf{0}$$
$$H_a : \boldsymbol{\alpha} \neq \mathbf{0},$$

where $\boldsymbol{\zeta}_{\tau-(h+1)}$ is a $k \times 1$ vector of variables contained in the information set $\boldsymbol{I}_{\tau-(h+1)}$, and the null and alternative hypotheses involve k parameters each. The forecasters are indexed by $i = 1, \ldots, N$, the target years by $\tau = 1, \ldots, T$, and the forecast horizons by $h = 1, \ldots, H$. For estimation, the data are sorted first by forecaster, then by target year, and lastly by forecast horizon. Under the joint hypothesis of forecast efficiency and quadratic loss, the optimal revision is uncorrelated with the variables contained in each forecaster's information set, $\boldsymbol{\zeta}_{\tau-(h+1)}$, and hence all of the components of $\boldsymbol{\alpha}$ should equal zero. Note that we are assuming the same quadratic loss function for all the forecasters (i.e., $\boldsymbol{\alpha}_i = \boldsymbol{\alpha}\ \forall i$).

Using Eq. (10) as a reference, we can perform a test of weak efficiency by incorporating the first two properties of optimal revisions in a single regression with $\boldsymbol{\zeta}_{\tau-(h+1)} = \left[1, \upsilon_{\tau,h+j}\right]$. In this sense, we are testing for both unbiasedness and a lack of autocorrelation in the forecast revision process. The regression is

$$\upsilon_{i,\tau,h} = \alpha_0 + \sum_{j=1}^{p} \alpha_j \upsilon_{i,\tau,h+j} + \epsilon_{1i,\tau,h}, \tag{11}$$
$$H_0 : \alpha_0 = \alpha_1 = \cdots = \alpha_p = 0.$$

In addition, as can be seen from property 3, optimal revisions incorporate publicly available information in an efficient manner, and therefore, they are revised exclusively in response to unexpected shocks. Thus, a test of strong efficiency can be performed by regressing the forecast revision on the candidate variables contained in each respondent's information set. Since the information set might differ across individuals and is not observable, any variable employed for evaluation purposes is rather arbitrary (Thomas, 1999). In order to avoid using variables which were not available to the respondents at the time when the forecasts were made, we use the monthly observed changes in the target variables as independent variables.

In the case of annual inflation, we use monthly inflation as the independent variable. We decided to include the second lag of the change in inflation (e.g., for the forecasts collected in April of each year, we include the change in inflation between January and February as part of their information set). These data are definitely published by the time the forecasts are collected, since the monthly inflation of the previous month is already available on the 10th day of each month (e.g., the value of inflation in March is published on April 10th).

For GDP growth, because actual values of GDP are not published on a monthly basis, we employ a coincident indicator of economic activity that includes among its components the indicator of global economic activity (IGAE), which serves as the monthly leading indicator of Mexican GDP, along with an indicator of industrial economic activity, the unemployment rate, total imports, an index of retail sales, and the number of people with social security. This indicator is published by INEGI around the second week of each month with a lag of three months (e.g., the value for January is published on the 12th of April). We decided to include the third lag of the change in this index to make sure that the information was available to all forecasters.

The strong efficiency regression is

$$\upsilon_{i,\tau,h} = \alpha'_1(x_{\tau-(h+1)} - x_{\tau-(h+2)}) + \epsilon_{2i,\tau,h}, \tag{12}$$
$$H_0 : \alpha'_1 = 0$$
$$H_a : \alpha'_1 \neq 0.$$

Notice that we can express the revisions in a vector, $\boldsymbol{\Upsilon}$, as follows:

$$\boldsymbol{\Upsilon}' = (\upsilon_{1,1,H-1}, \ldots, \upsilon_{1,1,1}, \upsilon_{1,2,H-1}, \ldots, \upsilon_{1,2,1}, \ldots, \upsilon_{1,T,H-1}, \ldots, \upsilon_{1,T,1}, \upsilon_{2,1,H-1}, \ldots, \upsilon_{N,T,H-1}).$$

3.4. Econometric considerations

We can provide consistent estimates of the relevant coefficients of Eqs. (11) and (12) for each individual respondent and for the pooled data using ordinary least squares (OLS). However, under the null hypothesis of efficiency, the error terms $\epsilon_{1i,\tau,h}$ and $\epsilon_{2i,\tau,h}$ will have special structures. Following Davies and Lahiri (1995), since the forecasts span a period of up to 24 months, with monthly revisions, the structure of the variance–covariance matrix of the error vector, $\boldsymbol{\Omega}$, should take into account: (i) different error variances across forecasters; and (ii) the correlations of contemporaneous revisions across individuals for the same target year. In addition, since, in any given month, forecasters will be revising their predictions for two consecutive years, consideration needs to be given to (iii) contemporaneous revisions for consecutive target years for each forecaster; and (iv) such revisions across forecasters.

The elements of $\boldsymbol{\Omega}$ can then be estimated from the OLS residuals $\widehat{\epsilon}_{1i,\tau,h}$ (or $\widehat{\epsilon}_{2i,\tau,h}$) by subtracting means and averaging across horizons and target years (Isiklar, Lahiri, & Loungani, 2006). We assume that all other covariances among $\epsilon_{1i,\tau,h}$ and $\epsilon_{2i,\tau,h}$ are zero. Using the general efficiency regression in Eq. (10):

- For the error variance of forecaster i, σ_i^2, we estimate:

$$\widehat{\sigma}_i^2 = \frac{1}{TH}\sum_{h=1}^{H}\sum_{t=1}^{T}\widehat{\epsilon}_{i,\tau,h}^2.$$

- The covariances across respondents, γ_{ij}, are computed as:

$$\widehat{\gamma}_{ij} = \frac{1}{TH}\sum_{h=1}^{H}\sum_{t=1}^{T}\widehat{\epsilon}_{i,\tau,h}\widehat{\epsilon}_{j,\tau,h}, \quad \forall i \neq j.$$

- Contemporaneous covariances for consecutive target years for each forecaster are estimated as:

$$\widehat{\omega}_i = \frac{1}{(T-1)\widetilde{H}}\sum_{h=1}^{\widetilde{H}}\sum_{t=1}^{T-1}\widehat{\epsilon}_{i,\tau,h}\widehat{\epsilon}_{i,\tau+1,h+12},$$

  where $\widetilde{H} < 12$.

Table 2
Evaluation of weak efficiency, inflation and GDP forecast revisions, 1995–2010.

Variables	Inflation	GDP
α	0.008	−0.045**
	(0.019)	(0.019)
$\upsilon_{\tau,h+1,i}$	0.010	0.003
	(0.021)	(0.027)
$\upsilon_{\tau,h+2,i}$	0.051***	0.158***
	(0.019)	(0.026)
Observations	6067	5994
Number of forecasters	41	41

** and *** denote statistical significance at the 5%, and 1% levels, respectively. GMM standard errors using the variance–covariance matrix given in the text are in parentheses.

- The contemporaneous covariances for consecutive target years across forecasters, s_{ij}, are computed as follows:

$$\widehat{s}_{ij} = \frac{1}{2(T-1)\widetilde{H}} \sum_{h=1}^{\widetilde{H}} \sum_{t=1}^{T-1} \widehat{\epsilon}_{i,\tau,h}\widehat{\epsilon}_{j,\tau+1,h+12} + \widehat{\epsilon}_{j,\tau,h}\widehat{\epsilon}_{i,\tau+1,h+12}, \quad \text{where } \widetilde{H} < 12.$$

The numbers of diagonal elements estimated, $\widehat{\sigma}_i^2$, and of individual covariances for the same target, $\widehat{\omega}_i$, equal the number of forecasters, N. The number of estimated covariances across forecasters for the same target, $\widehat{\gamma}_{ij}$, and for consecutive targets, $\widehat{s}_{ij}$, is $(N \times (N-1))/2$.

In order to obtain consistent standard errors for the coefficients of Eqs. (11) and (12), we calculate the general method of moments (GMM) covariance estimator, given by $(\boldsymbol{\zeta}'\boldsymbol{\zeta})^{-1}\boldsymbol{\zeta}'\widehat{\boldsymbol{\Omega}}\boldsymbol{\zeta}(\boldsymbol{\zeta}'\boldsymbol{\zeta})^{-1}$. The consistent GMM standard errors are then obtained as the square root of the diagonal elements of this covariance estimator.

4. Empirical results from efficiency tests

In this section, we present the empirical results of the weak and strong efficiency tests developed above. We focus on the pooled results containing the information on the panel data structure, and, where possible, discuss separate individual results obtained by using the relevant portion of the estimated variance–covariance matrix. These results should be treated with caution because of the low power of the tests when applied to small samples, and because the gaps in the individual time series are particularly problematic for estimating autocorrelations in the forecast revision process.

4.1. Weak efficiency

The results of the weak efficiency tests applied to the pooled forecast revisions are shown in Table 2, along with consistent standard errors obtained from the GMM covariance estimator. The second column presents the results for the weak-efficiency test with $p = 2$. That is, we are testing whether the forecast revisions at period $\tau - h$ are correlated with past revisions at periods $\tau - (h+1)$ and $\tau - (h+2)$.

First, these results indicate that, on average, SPF's respondents are unbiased for annual inflation forecast revisions, which is consistent with property 1 of an optimal forecast revision.

Second, the results for annual inflation show that the first-order autocorrelation coefficient in the regression in Eq. (11) is slightly positive, but not statistically different from zero at conventional significance levels, and that the second-order autocorrelation coefficient is significantly different from zero at the 1% level, suggesting an inefficient use of the available information on the part of the forecasters. The fact that α_2 is estimated to be positive can be interpreted as evidence of over-smoothing. That is, forecasters seem to maintain their initial forecasts for too long, which could be related to "anchoring", the common human tendency to rely too heavily on one piece of information when making decisions (Tversky & Kahneman, 1974).

The individual results for annual inflation show that six of the 41 forecasters analyzed (15%) are found to have biases in their revisions. Of these forecasters, five of the six biases are positive, suggesting that these forecasters systematically revised their inflation forecasts upward. Also, 22 of the 41 forecasts analyzed (54%) have significant levels of autocorrelation in their forecast revisions, with nine of the 22 (41%) showing positive serial correlation in both lags, three of them showing negative serial correlation in both lags (14%), and 10 of them (45%) showing different signs in their coefficient estimates for lagged revisions.

The efficiency tests for the inflation forecasts performed above for the full period 1995–2010 do not take into account potential changes in the estimated parameters, or in the structure of the variance–covariance matrix before and after the initiation of the inflation targeting regime (IT). The drastic change in inflation persistence that coincides with the implementation of IT in 2001, as documented by Chiquiar et al. (2010), renders especially pertinent an analysis of the forecast revision process before and after its implementation. To analyze the efficiency of inflation forecasts for these two sub-periods, we constructed a dummy variable ($IT_{\tau,h}$) that takes the value of one for every forecast reported after the year 2001, which is the year when IT was implemented by Banco de México. In this way, we have been able to estimate a weak efficiency test both before and after IT in the same regression, using an interaction model of the form:

$$\upsilon_{i,\tau,h} = \alpha_0 + \alpha_1\upsilon_{\tau,h+1,i} + \alpha_2\upsilon_{\tau,h+2,i} + \alpha_3 IT_{\tau,h} + \alpha_4(\upsilon_{\tau,h+1,i} \times IT_{\tau,h}) + \alpha_5(\upsilon_{\tau,h+1,i} \times IT_{\tau,h}) + \epsilon_{1i,\tau,h}.$$

Notice that if $IT_{\tau,h} = 0$, we obtain the weak efficiency tests from Eq. (11) for the sub-sample 1995–2001, whereas if $IT_{\tau,h} = 1$, the weak efficiency test for the period 2002–2010 can be written as

$$\upsilon_{i,\tau,h} = (\alpha_0 + \alpha_3) + (\alpha_1 + \alpha_4)\upsilon_{\tau,h+1,i} + (\alpha_2 + \alpha_5)\upsilon_{\tau,h+2,i} + \epsilon_{1i,\tau,h}.$$

The results of this exercise are presented in Table 3. Column 1 of this table estimates the efficiency tests by

Table 3
Evaluation of weak efficiency for inflation before and after inflation targeting.

Variables	All forecasters	Same forecasters
$\alpha(\Pr e - IT)$	−0.054	−0.057
	(0.043)	(0.042)
$\upsilon_{\tau,h+1,i}(\Pr e - IT)$	0.012	0.011
	(0.023)	(0.023)
$\upsilon_{\tau,h+2,i}(\Pr e - IT)$	0.051**	0.048**
	(0.021)	(0.020)
$IT_{\tau,h}$	0.092*	0.095**
	(0.047)	(0.048)
$\upsilon_{\tau,h+1,i} \times IT_{\tau,h}$	−0.044	−0.043
	(0.046)	(0.051)
$\upsilon_{\tau,h+2,i} \times IT_{\tau,h}$	−0.021	−0.025
	(0.044)	(0.050)
$\alpha(Post - IT)$	0.039**	0.038*
	(0.019)	(0.023)
$\upsilon_{\tau,h+1,i}(Post - IT)$	−0.032	−0.033
	(0.040)	(0.046)
$\upsilon_{\tau,h+2,i}(Post - IT)$	0.030	0.024
	(0.039)	(0.045)
Observations	6067	5005
Number of forecasters	41	31

* and ** denote statistical significance at the 10% and 5% levels, respectively. GMM standard errors using the variance–covariance matrix given in the text are in parentheses. $IT_{\tau,h}$ is a dummy variable that takes the value of 1 whenever a forecast is reported after 2001. 'Same forecasters' refers to a specification using only those individuals with at least 10 forecasts before inflation targeting.

pooling all of the forecasters available in either of the subsamples, while the second column of these tables presents the results using only those forecasters who predicted inflation both before and after the implementation of IT.[10] By restricting the sample to the same forecasters before and after IT, we reduce the problem of having individual forecasters entering and exiting the survey frequently.

The weak efficiency tests indicate that the picture we obtain for the period before IT (1995–2001) is very similar to the results for the whole sample, with forecasters "anchoring" to their initial forecasts and smoothing their revisions as they receive more information. However, the second-order autocorrelation in the revision process is reduced greatly when we look at the period after IT (2002–2010), and it is not statistically significant at conventional levels, which suggests that, on average, the forecasts reported after 2001 were more efficient in incorporating available information.[11]

Notice that even though the evidence of over-smoothing appears to have diminished under IT, the forecasters systematically increase their forecasts in this period, which is consistent with the evidence for the period under IT, presented in Table 1 and Fig. 1, that forecasters tended to initially under-predict the annual inflation at long horizons, then reduced this bias in subsequent revisions.

The panel results for GDP growth forecast revisions follow the same pattern as for inflation in the period 1995–2009. As in the previous case, we are able to reject the weak-efficiency hypothesis, although in this case, we also find a negative and significant constant bias, which implies that, on average, these forecasters were systematically decreasing their forecasts for annual GDP growth. This is consistent with the evidence presented in Fig. 2, that the initial forecasts tend to over-predict GDP growth. Hence, the systematic bias in the revisions may be an attempt to reduce the bias in shorter-horizon forecasts.

The individual results for annual GDP growth show that five of the 41 forecasters analyzed (12%) are found to have significant biases in their revisions, all of them negative, suggesting that these forecasters systematically revised their GDP growth forecasts downwards. Also, 28 of the 41 forecasts (54%) of the forecasters analyzed have significant levels of autocorrelation in their forecast revisions, with 17 of them (61%) presenting positive serial correlations in both lags, two of them presenting negative serial correlations in both lags (7%), and nine of them (32%) presenting different signs in their coefficient estimates for lagged revisions.

Table 4
Evaluation of strong efficiency, inflation and GDP forecast revisions, 1995–2010.

Variables	Inflation	GDP
γ	0.026	−0.043***
	(0.023)	(0.017)
$x_{t,h+1} - x_{t,h+2}$	0.257***	0.899***
	(0.058)	(0.109)
Observations	7635	7538
Number of forecasters	41	41

*** denotes statistical significance at the 1% level. GMM standard errors using the variance–covariance matrix given in the text are in parentheses.

4.2. Strong efficiency

The results of testing the strong efficiency property with consistent GMM standard errors for both inflation and GDP growth are presented in Table 4. We include a constant in the regression in order to allow for the presence of a systematic bias in the forecast revision process. We also include a $MA(1)$ term in the error process to capture any possible autocorrelation that might bias the estimated coefficients' standard errors.

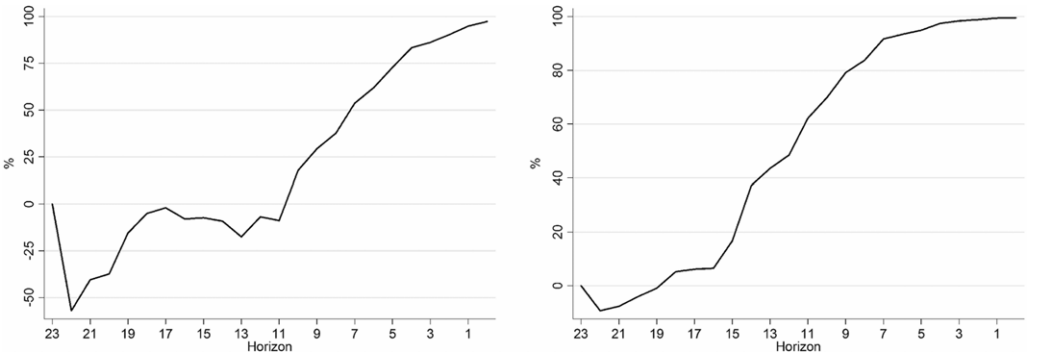
The aggregate results for the inflation forecasts reject the strong efficiency hypothesis at the 1% significance level. The positive sign of the estimated coefficient indicates that increases in past monthly inflation levels are associated with positive monthly revisions in the annual forecasts. In particular, a 1% increase in the monthly inflation news induces an upward change in the monthly revision process of 0.27%. In principle, this evidence suggests that annual inflation forecasts could be improved by using information contained in the past evolution of monthly inflation.

At the individual level, we find that 11 of the 41 forecasters (27%) did not incorporate past values of monthly inflation efficiently, as they are positively and significantly correlated with their forecast revisions.

[10] In this case, the criterion was that forecasters must have at least 10 forecasts before 2002.

[11] For the period under IT (2002–2010), the second-order autocorrelation is calculated as $\alpha_2 + \alpha_5$, with the standard error given by

$$s.e.(\alpha_2 + \alpha_5) = \sqrt{var(\widehat{\alpha}_2) + var(\widehat{\alpha}_5) + 2cov(\widehat{\alpha}_2, \widehat{\alpha}_5)}.$$



Inflation forecasts GDP Growth Forecasts

* The measure of information content is $pred_{h,23} = \left(1 - \frac{MSE_h}{MSE_{23}}\right) 100$, as per Diebold and Kilian (2001); it gives the improvement in the forecast (over the 24-month-ahead prediction) as the horizon decreases.

** For the inflation forecasts, we use information over the period 1999–2009, when 23 horizons are available for each event.

*** For the GDP growth forecasts, we use information over the period 2000–2009, when 23 horizons are available for each event.

Fig. 4. Information content of forecasts over different horizons.

Table 5
Evaluation of weak efficiency for inflation before and after inflation targeting.

Variables	All forecasters	Same forecasters
$\gamma(\Pr e - IT)$	0.009	0.008
	(0.049)	(0.049)
$x_{t,h+1} - x_{t,h+2}(\Pr e - IT)$	0.456***	0.474***
	(0.089)	(0.091)
$IT_{\tau,h}$	0.027	0.026
	(0.054)	(0.056)
$(x_{t,h+1} - x_{t,h+2}) \times IT_{\tau,h}$	−0.422***	−0.439***
	(0.113)	(0.123)
$\gamma(Post - IT)$	0.035	0.035
	(0.023)	(0.028)
$x_{t,h+1} - x_{t,h+2}(Post - IT)$	0.034	0.035
	(0.070)	(0.083)
Observations	7635	6531
Number of forecasters	41	31

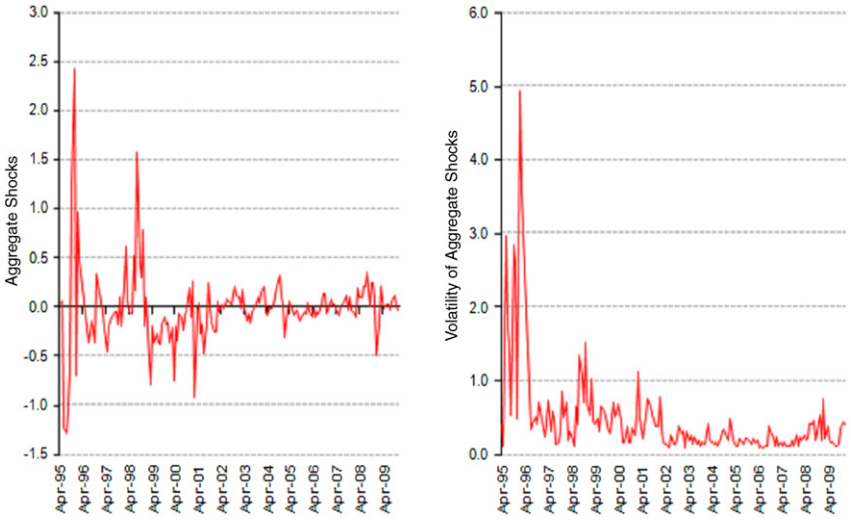
*** denotes statistical significance at the 1% level. GMM standard errors using the variance–covariance matrix given in the text are in parentheses. $IT_{\tau,h}$ is a dummy variable that takes the value of 1 whenever a forecast is reported after 2001. 'Same forecasters' refers to a specification using only those individuals with at least 10 forecasts before inflation targeting.

Table 5 presents the strong efficiency tests for the inflation forecasts when we analyze the forecast revisions before and after the implementation of inflation targeting. As can be seen, the rejection of the strong efficiency hypothesis for the whole period under study seems to be driven by the period before the implementation of IT, when forecasters failed to include the available information on previous values of monthly inflation when revising their annual forecasts. Once we look at the post-IT period, we see that we cannot reject the null of strong efficiency, as past values of monthly inflation do not seem to be systematically correlated with forecast revisions. As the second column of Table 5 indicates, these results are robust to using the same subset of forecasters before and after the introduction of IT.

In general terms, the inefficiencies in the forecast revision process seem to have diminished under IT. This result is consistent with the fact that inflation in Mexico has behaved as a stationary process, which can be easier to predict using simple time series models, for example.[12]

Strong efficiency is also strongly rejected in the case of annual GDP growth forecasts. The estimated coefficients presented in the third column of Table 4 show that, when predicting GDP growth, not only are forecasters biased downwards in their revisions, they also fail to use all of the valuable information on past values of the indicator of economic activity, which is publicly available at the time when they are reporting their forecasts.

[12] A similar result is found for efficiency tests of fixed-horizon forecasts from Banco de México's SPF (Capistrán & López-Moctezuma, 2010).



* Estimates of the aggregate shocks are obtained by averaging the inefficiency-corrected shocks each month, as detailed in the paper.

** Estimates of the volatility of aggregate shocks are estimated by taking the interquartile range on the inefficiency-corrected shocks,as detailed in the paper.

Fig. 5. Aggregate shocks to expected inflation.

5. Estimates of the monthly aggregate shocks

In this section, we provide estimates of the aggregate shocks that might have affected the SPF's respondents homogeneously at each point in time. It is important to note that, since the estimates obtained through Eq. (9) can only be interpreted as pure aggregate shocks under the joint hypothesis of optimality and quadratic loss, in order to extract the perceived impact of monthly aggregate shocks, we estimated the residuals of Eq. (11) as "inefficiency-corrected" shocks, which can be viewed as ideal revisions if the forecasters were efficient:

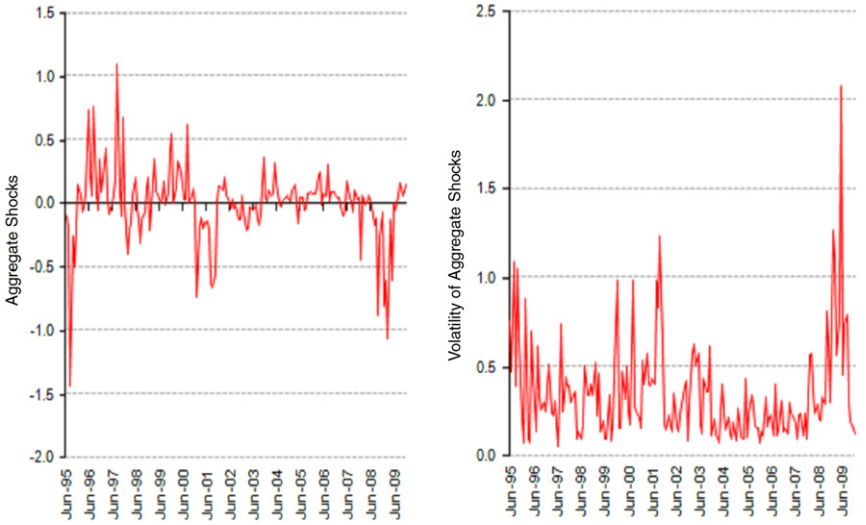
$$\upsilon^{*}_{i,\tau,h} = \upsilon_{i,\tau,h} - (\widehat{\alpha}_0 + \widehat{\alpha}_1 \upsilon_{i,\tau,h+1} + \widehat{\alpha}_2 \upsilon_{i,\tau,h+2}).$$

Then, we use Eq. (9) to calculate time series estimates of the impact of aggregate macroeconomic shocks which affect the expectation formation process of annual inflation and GDP growth forecasts.[13] Those corresponding to inflation are plotted in Fig. 5, while those corresponding to GDP growth can be found in Fig. 6. Positive values of these shocks imply an increase in the value of the target variable, while negative values refer to a decrease in the value of the target variable. In this sense, positive shocks can be interpreted as "bad news" for inflation forecasts and "good news" for GDP growth forecasts, and vice versa. The dates on the horizontal axis give the month of the survey that collected the revised forecast, indicating the timing of the new information that may have influenced the revision. In both figures, the left panel plots the aggregate shocks and the right panel plots the volatility of those shocks, obtained as the interquartile range of forecast revisions across respondents. The larger the volatility of the aggregate shock in a given month, the larger the disagreement among forecasters as to the effect of that month's news on the target variable. This could be seen as a proxy for the overall uncertainty of forecasters regarding the impact of news.

The left panel of Fig. 5 shows that the largest shocks to inflation expectations in the sample under study happened in October and November of 1995, when the SPF respondents received two consecutive "bad news" shocks, of 1.2% and 2.4% respectively. These "bad news" shocks to aggregated inflation expectations can be regarded as a consequence of the Tequila crisis that started in December 1994, in which the sudden exchange rate devaluation triggered a new surge of inflation and a considerable drop in output during 1995, as internal adjustments were needed in order to absorb the impact of the rundown of international reserves and the posterior suspension of access to external savings (Gil-Díaz & Carstens, 1996). In order of magnitude, the next important shock to inflation expectations occurred in August 1998, when SPF's respondents revised their inflation expectations upwards by 1.58%. As was the case in the economic crisis of 1995, since the sign of this shock is positive, it can be interpreted as bad news for inflation (i.e., an increase in inflation). This shock could be related to the turbulence experienced by Latin American countries during this year as a consequence of the financial crises which originated in Asia and Russia in mid-1997. It can also be seen that, on average, the aggregate shocks between 1999 and 2001 are negative, implying good news for inflation. These "good news" shocks coincide with a series of disinflation efforts on the part of Banco de México, and also with the transition to an inflation-targeting regime that culminated with its implementation in 2001.

With respect to the volatility of the aggregate shocks to the expected inflation, as presented in the right panel of Fig. 5, it is worth noting the high volatility before 2001, in

[13] We thank an anonymous reviewer for suggesting this approach to estimating the monthly aggregate shocks.



* Estimates of the aggregate shocks are obtained by averaging the inefficiency-corrected shocks each month, as detailed in the paper.

** Estimates of the volatility of the aggregate shocks are estimated by taking the interquartile range of the inefficiency-corrected shocks, as detailed in the paper.

Fig. 6. Aggregate shocks to expected GDP growth.

periods of both bad and good news, and the low volatility following the implementation of inflation-targeting. In fact, the average interquartile range of the current year aggregate shocks from April 1995 to January 2001 is 0.75%, whereas from February 2001 until December 2009 this figure is 0.26%.

The left panel of Fig. 6 displays the sequence of monthly aggregate shocks available for GDP growth expectations for the current and following year forecasts. Consistent with the effects of the Tequila crisis on inflation expectations, the estimates of aggregate shocks to GDP growth forecasts also indicate one extreme negative ("bad news") shock on August 1995 of around 1.44%. In the same line, a series of "bad news" shocks to GDP growth expectations can be found in 1998. As in the case of inflation expectations, these are related to the effects of the Asian and Russian crises. Another series of bad news shocks to GDP growth forecast revisions can also be found from January to October of 2001, with a cumulative negative revision of 3.59%, which coincides with the recession in Mexico between 2001 and 2002, which in turn is related to the global economic downturn of the same years, particularly in the United States in the aftermath of the 9/11 terrorist attacks. In the last part of the sample, one can see a series of increasing bad news shocks appearing from January 2008 until the first half of 2009, related to the recent global financial crisis and the recession in the United States that started in December 2007. The volatility of the aggregate shocks to GDP growth forecast revisions indicates high levels of volatility for most of the sample, except for a relatively brief period of stability between 2004 and 2006. As a consequence of the most recent recession, the volatility of the aggregate shocks to the GDP growth forecasts has been increasing steadily in the last part of the sample.

6. Discussion

One important difference between survey forecasts and pure model-based forecasts is that the former typically include the subjective views (i.e., extra-sample information) of the experts that participate in the survey. This extra information, or expertise, may or may not help to make the forecasts more accurate than those of other forecasting methods, given a certain evaluation metric. However, before evaluating the relative performances of survey forecasts, it is necessary to determine whether or not the judgement put into these forecasts has induced systematic inefficiencies that may reduce their usefulness.

In this paper, we have evaluated the fixed-event forecasts of inflation and GDP growth from the respondents to the Banco de México's Survey of Professional Forecasters. As in other countries, the forecasters seem to start by giving forecasts which are close to what they consider to be the unconditional mean of the variable of interest, 24 months before the realization of the actual value. However, these assessments appear to be rather too rosy, both for inflation and for GDP growth. After the initial forecasts, news seems to start to be incorporated around 12 months before the realization of the actual end-of-the-year inflation, and about 16 months before in the case of GDP growth.

We exploit the fixed-event forecast structure of our data in order to study the efficiency of the forecast revision process. Tests of weak efficiency based on pooled estimations show evidence of over-smoothing (or anchoring) of the forecasts, in the sense that the revisions exhibit positive serial correlations. Hence, forecasters tend to hold to their views for too long.

With respect to strong efficiency, forecasters do not seem to systematically incorporate the information in

either past monthly inflation or the coincident indicator of economic activity when predicting inflation and GDP growth. In the case of inflation forecasts, though, this inefficient use of the available information appears to have diminished under inflation targeting.

We need to stress that the empirical results we have presented may be subject to change, especially in the case of inflation, given the relatively short duration of the SPF under an inflation targeting regime, and the fact that forecasters may be adjusting to the new regime gradually.

One important issue to point out is that the empirical assessment of efficiency implemented in this paper assumes that forecasters only care about deviations in their predictions with respect to the actual value of the target variable. However, professional forecasters might have other things to consider when computing their forecasts, as the forecasts they produce are important for decisions on monetary and fiscal policies, portfolio allocations, wage negotiations, etc. For example, Batchelor and Dua (1992) provide a plausible explanation of the forecasting "smoothness" displayed by perfectly rational forecasters, namely that forecasters are likely to take into account the fact that their clients might mistrust forecasters who make frequent revisions to forecasts, so that in this context, a smooth forecast evolution implies stability, with stable forecasts not being overly sensitive to new information.

The different incentives faced by the forecasters can also be consistent with the existence of other loss functions which attach different weights to the costs of over- and under-predicting the variable of interest. These asymmetric loss functions have been studied previously in the literature (Capistrán, 2008; Elliott, Komunjer, & Timmermann, 2005; Patton & Timmermann, 2007). In fact, the assumption of the loss function is crucial in determining whether the forecasters analyzed can be labeled efficient or not. As Patton and Timmermann (2007) demonstrated, none of the properties which are traditionally associated with optimal forecasts under a quadratic loss hold with asymmetric loss. The same conclusion arises when we deal with the properties of optimal revisions. In addition, it can also be shown that, under certain asymmetric loss functions, the forecast revision process includes an optimal bias, depending on the degree of asymmetry and on a measure of the uncertainty surrounding the target variable.

Hence, the inefficiencies found in the forecasts from the SPF are only so with respect to a particular cost function, the quadratic one. However, this does not imply that the forecasts cannot be useful to a decision-maker with a cost function of this type. As was shown by Bentancor and Pincheira (2010) and Capistrán and Timmermann (2009b), a user of the forecasts can take advantage of systematic inefficiencies, such as the bias and the excess of autocorrelation, to built new and more accurate forecasts. In this manner, the empirical results presented in this paper suggest that some gains may be achieved in the actual forecasts of the SPF from putting more weight on variables which are already at the disposal of the professional forecasters. Specifically, one important lesson from this analysis is that even the past performance of the consensus forecast could be employed to refine predictions.

In addition to the efficiency tests, we also estimated a measure of the aggregate monthly shocks that have affected inflation and GDP growth. These are shocks that had an impact on the forecasts, producing forecast errors, but for which the forecasters should not be held accountable. The estimated shocks appear to be related to specific periods of macroeconomic uncertainty. For instance, for inflation, we find a declining impact of the news at longer forecast horizons, and marked effects of the latest disinflation efforts (as good news shocks) and of inflation targeting (as reduced volatility). The aggregate shocks to the expected GDP growth show a clear effect of the business cycle, and of downturns in particular, and a high volatility overall.

Future research should consider the extent to which the inefficiencies found here can be rationalized by other loss functions, as per Capistrán and Timmermann (2009a). In the same vein, it is necessary to check what proportion of the macroeconomic aggregate shocks provided in this document are driven by the assumption of quadratic loss.

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